

Notes: Zwick & Mahon (AER, 2017)

Tax Policy and Heterogeneous Investment Behavior

Contents

1	Big picture	2
2	Data and measurement	2
2.1	Tax-return data and sample construction	2
2.2	Investment and firm variables	2
2.3	Policy variable: present value of depreciation deductions	3
2.4	Tax status and immediacy	3
3	Models and empirical specifications	3
3.1	Economic effect of bonus depreciation	3
3.2	Baseline difference-in-differences regression	3
3.3	Robustness and placebo specifications	4
3.4	Substitution and external-finance regressions	4
3.5	Heterogeneity by constraints	4
3.6	Tax-position interaction	4
3.7	Weighted elasticity	5
4	Identification and empirical design	5
4.1	What identifies the bonus effect	5
4.2	Why the design is credible	5
4.3	Why the tax-position tests matter	5
4.4	What the paper cannot fully prove	5
5	Tables and figures	6
5.1	Table 1 + Table 2: depreciation mechanics and sample statistics	6
5.2	Figure 1 + Table 3: baseline investment response	6
5.3	Table 4 + Figure 2: robustness and depreciation-kink salience	9
5.4	Table 5 + Table 6: substitution, financing, and constrained firms	9
5.5	Figure 3 + Table 7: tax position and immediacy	10
5.6	Figure 4: comparison to past work and firm-size heterogeneity	11
6	Short summary	12
7	Conclusion	12
7.1	Core conclusion	12
7.2	Economic intuition	13
7.3	Takeaway	13

1 Big picture

- **Question.** How much do temporary investment tax incentives affect corporate equipment investment, and which firms respond most strongly?
- **Policy shock.** The paper studies bonus depreciation, which accelerates depreciation deductions for eligible equipment. It changes the timing of tax deductions, not the total amount deducted.
- **Core identification.** Industries differ in the useful lives of the equipment they typically buy. Bonus depreciation is more valuable for long-duration equipment because moving future deductions to the present creates a larger present-value gain.
- **Data advantage.** The paper uses IRS corporate tax return data, including public and private firms. The main bonus sample has 820,769 firm-year observations for 128,151 firms.
- **Baseline effect.** Bonus depreciation raises eligible investment relative to less exposed industries by about 10.4% in 2001–2004 and 16.9% in 2008–2010.
- **Heterogeneity by size and constraints.** Small firms respond about 95% more than big firms. Non-dividend payers and low-cash firms also respond more strongly.
- **Tax-position result.** Firms respond strongly when the policy immediately reduces current taxes. Firms with tax losses, which can only carry deductions forward, respond weakly.
- **Section 179 evidence.** The paper documents sharp bunching at Section 179 expensing kinks. Firms bunch when the deduction immediately lowers their current tax bill, not when the benefit is delayed.
- **Economic takeaway.** Investment tax policy is not a uniform user-cost shock. Its effects are amplified by liquidity, financing frictions, or fixed adjustment costs, and accurate macro-relevant elasticities require accounting for heterogeneous firm responses.

2 Data and measurement

2.1 Tax-return data and sample construction

- The data come from the IRS Statistics of Income corporate tax return samples, which are stratified by total assets, proceeds, and form type.
- The raw panel covers 1993–2010 and contains 1.84 million firm-years for 355,000 distinct firms.
- The paper focuses on C and S corporations with substantial business activity and an attached depreciation form.
- The main bonus analysis excludes firms potentially affected by Section 179 and keeps firms with average eligible investment above \$100,000 in positive-investment years.
- Final main bonus sample: 820,769 observations for 128,151 firms.

Interpretation. The SOI data are crucial because they include small and private firms, not just large public Compustat firms. This is what lets the paper estimate heterogeneous responses across the firm-size distribution.

2.2 Investment and firm variables

- **Eligible investment** is equipment investment reported on the depreciation form for which bonus depreciation and Section 179 can apply.
- The main intensive-margin outcome is $\log(I_{it})$, where I_{it} is eligible investment.
- The extensive-margin outcome is the industry-level log odds of positive eligible investment.
- Alternative outcomes include investment divided by lagged capital stock, log structures investment, log capital stock growth, log debt growth, log rent growth, log payroll growth, and bonus take-up.
- Firm controls include assets, sales, profit margin, firm age, cash flow, debt, liquid assets, payroll, rents, and interest.

Interpretation. The main investment outcome is log investment because investment is highly skewed: the mean eligible investment observation is about \$6.8 million, but the median is only about \$370,000.

2.3 Policy variable: present value of depreciation deductions

For a dollar of investment, the normal depreciation value is

$$z^0 = D_0 + \sum_{t=1}^T \frac{1}{(1+r)^t} D_t, \quad (1)$$

where D_t is the allowed deduction per dollar of investment in year t , T is the asset class life, and r is the risk-adjusted discount rate.

With bonus depreciation, the present value becomes

$$z = \theta + (1 - \theta)z^0, \quad (2)$$

where θ is the bonus fraction immediately deductible.

The empirical policy variable is $z_{N,t}$, the four-digit NAICS industry present value of a dollar of eligible investment, adjusted in bonus years by the statutory bonus rate.

Interpretation. The cross-sectional variation comes from industry investment composition: firms in industries that normally buy longer-lived assets receive larger present-value tax benefits from bonus depreciation.

2.4 Tax status and immediacy

- A firm is in taxable position when net income before depreciation is positive.
- Taxable firms get current cash-flow benefits from extra depreciation deductions.
- Loss firms can carry deductions forward, so the same statutory incentive has weaker immediate cash-flow value.
- The paper also uses lagged loss carryforward stocks to measure the availability of alternative tax shields.

Interpretation. This variable is central for the mechanism. If only current-tax firms respond, then the effect is not simply a neoclassical present-value user-cost response; it also reflects liquidity or high effective discount rates.

3 Models and empirical specifications

3.1 Economic effect of bonus depreciation

Bonus changes the after-tax cost of an investment by changing the timing of deductions. If τ is the corporate tax rate, the net-of-tax investment price is proportional to

$$1 - \tau z. \quad (3)$$

A rise in z lowers the effective price of eligible investment.

Interpretation. The paper's main coefficient can be read as a semi-elasticity of investment with respect to the present value of depreciation deductions.

3.2 Baseline difference-in-differences regression

The main regression is

$$f(I_{it}, K_{i,t-1}) = \alpha_i + \beta g(z_{N,t}) + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (4)$$

- i indexes firms and t indexes years.
- $z_{N,t}$ is the industry-year policy exposure.
- α_i are firm fixed effects; δ_t are year fixed effects.
- X_{it} are firm-level controls in some specifications.

- For the intensive margin, $f(I, K) = \log(I_{it})$ and $g(z) = z_{N,t}$.
- For the extensive margin, $f(I, K)$ is the industry log odds of positive investment.
- For the tax-term model, $f(I, K) = I_{it}/K_{i,t-1}$ and $g(z) = (1 - \tau z_{N,t})/(1 - \tau)$.

Interpretation. The design compares firms in the same year but in industries receiving different bonus-induced price cuts. Firm fixed effects absorb permanent firm differences; year fixed effects absorb aggregate recessions and nationwide policy changes.

3.3 Robustness and placebo specifications

The robustness regressions keep the same basic structure but change the dependent variable:

$$Y_{it} = \alpha_i + \beta z_{N,t} + \delta_t + \varepsilon_{it}. \quad (5)$$

The paper uses three key checks:

- replace eligible equipment investment with structures investment, which is generally ineligible for bonus;
- replace gross eligible investment with net capital-stock growth;
- replace investment with an indicator for whether the firm reports bonus depreciation take-up.

Interpretation. The structures placebo asks whether treated industries were simply booming. The take-up test asks whether the tax data show firms actually claiming the policy where the estimated exposure is high.

3.4 Substitution and external-finance regressions

For margins such as rents, payroll, debt, and dividend-paying status, the paper estimates

$$Y_{it} = \alpha_i + \beta z_{N,t} + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (6)$$

Interpretation. These regressions ask whether bonus merely shifts spending across categories or whether it coincides with financing responses. The evidence points to higher debt and lower payout, rather than simple substitution away from labor or leasing.

3.5 Heterogeneity by constraints

The paper splits the baseline intensive-margin model by ex ante constraints:

- small versus big firms, using pre-policy sales deciles;
- dividend payers versus nonpayers;
- high versus low lagged cash, residualized against size, industry, time, and corporate form.

Interpretation. If financial frictions amplify the policy effect, constrained firms should have larger β coefficients. This is what the paper finds.

3.6 Tax-position interaction

A simplified version of the tax-position regression is

$$\log(I_{it}) = \alpha_i + \beta_T(\text{Taxable}_{it} \times z_{N,t}) + \beta_N z_{N,t} + \gamma X_{it} + \delta_t + \varepsilon_{it}. \quad (7)$$

Interpretation. β_T measures whether current-taxable firms respond more to bonus than loss firms. A large positive interaction means immediate cash-flow benefits are central.

3.7 Weighted elasticity

Because investment is concentrated among large firms, the paper computes a size-bin weighted elasticity:

$$\eta^W = \sum_{s=1}^S \omega_s \eta_s, \quad (8)$$

where η_s is the elasticity for size bin s and ω_s is that bin's share of aggregate eligible investment.

Interpretation. Equal-weighted estimates emphasize the many small firms in the tax data. Investment-weighted estimates ask how much aggregate investment moves.

4 Identification and empirical design

4.1 What identifies the bonus effect

- The treatment is not simply the presence of a bonus year. Everyone is exposed to the calendar-year policy.
- Identification uses differential exposure across industries caused by asset lives.
- Industries with long-duration equipment receive a larger present-value increase in depreciation allowances.
- The key assumption is that long-duration and short-duration industries would have followed similar investment trends absent bonus depreciation.

Interpretation. The design is a continuous-treatment difference-in-differences design with staggered policy intensity across industries.

4.2 Why the design is credible

- Figure 1 shows parallel pretrends and a break during bonus years in both the 2001–2004 and 2008–2010 episodes.
- The effects are stable after adding cash-flow controls, public-company industry Q , firm controls, and flexible industry trends.
- Placebo tests show no effect on structures investment, which is largely ineligible for bonus.
- Bonus take-up is higher in more exposed industries.
- Past recessions without investment stimulus do not generate similar positive placebo effects.

Interpretation. These checks address the main threat: that treated industries were hit by unrelated industry-specific shocks exactly when bonus depreciation was enacted.

4.3 Why the tax-position tests matter

- If firms respond because of standard present-value incentives alone, loss firms should still value future deductions.
- If firms are liquidity constrained or use high effective discount rates, only current-tax benefits should matter strongly.
- The paper finds the latter: taxable firms respond; nontaxable firms respond weakly.

Interpretation. This result is the strongest mechanism evidence that immediacy matters.

4.4 What the paper cannot fully prove

- The cross-sectional design does not estimate the general-equilibrium aggregate effect of bonus depreciation.
- Industry-by-year policy exposure may still be correlated with unobserved shocks, though the placebo and robustness tests reduce this concern.

- The heterogeneity evidence supports financial frictions or fixed adjustment costs, but it does not fully distinguish between them.
- Tax-return data are rich but anonymized, so some firm-level investment categories and financing details remain unavailable.

Interpretation. The paper’s contribution is a credible and unusually broad micro estimate of how firms respond, not a complete macro multiplier calculation.

5 Tables and figures

5.1 Table 1 + Table 2: depreciation mechanics and sample statistics

Read:

- Table 1 shows why timing matters: with 50% bonus on a \$1 million computer investment, the first-year deduction rises from \$200,000 to \$600,000, but total deductions remain \$1 million.
- Table 2 shows a skewed sample: mean eligible investment is \$6.79 million, while the median is \$367,590.
- Average $z_{N,t}$ is 0.907 at the firm level, and the industry policy variable varies meaningfully across industries and bonus years.

Economic message: Bonus depreciation is a timing subsidy. It is most valuable when a firm buys assets whose normal deductions arrive far in the future.

Year:	0	1	2	3	4	5	Total
<i>Normal depreciation</i>							
Deductions (000s)	200	320	192	115	115	58	1,000
Tax benefit ($\tau = 35$ percent)	70	112	67.2	40.3	40.3	20.2	350
<i>Bonus depreciation (50 percent)</i>							
Deductions (000s)	600	160	96	57.5	57.5	29	1,000
Tax benefit ($\tau = 35$ percent)	210	56	33.6	20.2	20.2	10	350

Notes: This table displays year-by-year deductions and tax benefits for a \$1 million investment in computers, a five-year item, depreciable according to the Modified Accelerated Cost Recovery System (MACRS). The top schedule applies during normal times. It reflects a half-year convention for the purchase year and a 200 percent declining balance method (2× straight line until straight line is greater). The bottom schedule applies when 50 percent bonus depreciation is available.

Source: Authors’ calculations. See IRS publication 946 for the recovery periods and schedules applying to other class lives (<https://www.irs.gov/uac/about-publication-946>).

	Mean	P10	Median	P90	Count
<i>Investment variables</i>					
Investment (000s)	6,786.87	0.81	367.59	5,900.17	818,576
log (investment)	6.27	4.10	6.14	8.81	735,341
Investment/lagged capital stock	0.10	0.00	0.05	0.27	637,243
Δ log (capital stock)	0.08	−0.05	0.05	0.33	637,278
Δ log (odds ratio) _{it}	1.28	0.54	1.34	2.05	818,107
<i>Other outcome variables</i>					
Δ log (debt)	0.04	−0.37	0.03	0.56	642,546
Δ log (rent)	0.08	−0.38	0.04	0.66	574,305
Δ log (wage compensation)	0.06	−0.21	0.05	0.40	624,918
log (structures investment)	5.02	2.13	4.98	8.10	389,232
<i>Policy variables (firm)</i>					
$z_{N,t}$	0.907	0.874	0.894	0.946	818,576
<i>Policy variables (industry)</i>					
z_N	0.881	0.866	0.885	0.895	317
$z_{N,t}$	0.903	0.870	0.892	0.945	5,125
<i>Characteristics</i>					
Assets (000s)	403,597.2	3,267.96	24,274.82	327,301.6	818,576
Sales (000s)	180,423.8	834.65	25,920.92	234,076.1	818,576
Capital stock (000s)	89,977.09	932.00	7,214.53	80,122.69	818,576
Net income before depreciation (000s)	15,392.59	−2,397.92	1,474.65	17,174.55	818,576
Profit margin	0.17	−0.07	0.05	0.68	777,968
Wage compensation (000s)	26,826.36	372.09	4,199.88	38,526.46	818,576
Cash flow/lagged capital stock	0.05	−0.09	0.03	0.26	647,617

Notes: This table presents summary statistics for analysis of bonus depreciation. To preserve taxpayer anonymity, “percentiles” are presented as means of all observations in the $(P - 1, P + 1)$ th percentiles. Investment is bonus eligible equipment investment. $z_{N,t}$ is the weighted present value for a dollar of eligible investment expense at the four-digit NAICS level, with weights computed using shares of investment in each eligible category. Statistics are presented at both the firm-by-year and industry-by-year level. z_N is the weighted present value for a dollar of eligible investment in non-bonus periods for the cross section of industries in the sample. The odds ratio is defined at the four-digit NAICS level as the fraction of firms with positive investment divided by the fraction with zero investment. Cash flow is net income before depreciation after taxes paid. Ratios are censored at the 1 percent level. Online Appendix Table B.1 presents more detailed investment statistics, allowing comparison of our sample to past work.

Source: Authors’ calculations

Table 1 and Table 2.

5.2 Figure 1 + Table 3: baseline investment response

Read:

- Figure 1 shows that long-duration industries and short-duration industries track each other before bonus, then separate during bonus years.
- Table 3 estimates a baseline intensive-margin coefficient of 3.69 on $z_{N,t}$.

- The extensive-margin coefficient is 3.79, and the tax-term elasticity is -1.60 .
- The coefficient implies investment increases of about 17.7 log points in the early bonus period and 28.8 log points in the later bonus period, using the average bonus-induced changes in $z_{N,t}$.

Economic message: The main result is visually and statistically clear: industries that receive larger bonus-induced tax-price cuts raise eligible investment more.

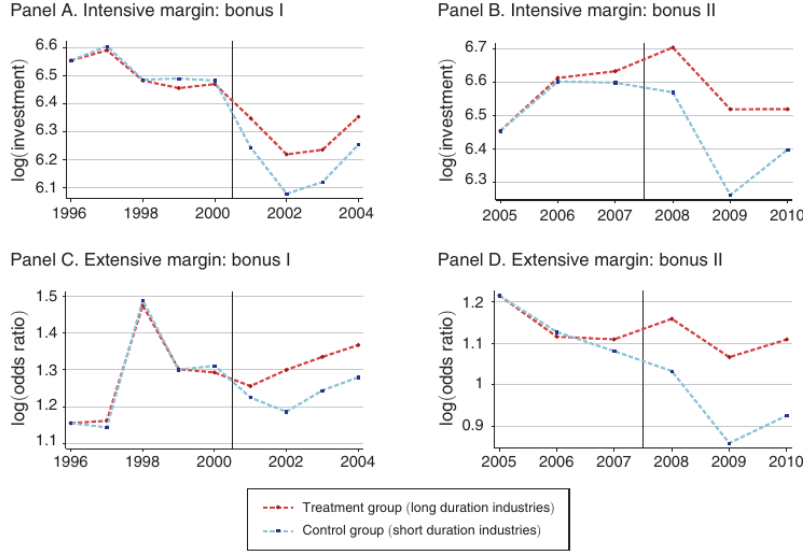


FIGURE 1. CALENDAR DIFFERENCE-IN-DIFFERENCES

Notes: The top graphs plot the average logarithm of eligible investment over time for groups sorted according to their industry-based treatment intensity. Treatment intensity depends on the average duration of investment, with long-duration industries (treatment groups) seeing a larger average price cut due to bonus than short-duration industries (control groups). The bottom graphs plot the industry-level log odds ratio for the probability of positive eligi-

TABLE 3—INVESTMENT RESPONSE TO BONUS DEPRECIATION

	Intensive margin: LHS variable is log(investment)					
	(1)	(2)	(3)	(4)	(5)	(6)
$z_{N,t}$	3.69 (0.53)	3.78 (0.57)	3.07 (0.69)	3.02 (0.81)	3.73 (0.70)	4.69 (0.62)
$CF_{it}/K_{i,t-1}$		0.44 (0.016)				
Observations	735,341	580,422	514,035	221,306	585,914	722,262
Clusters (firms)	128,001	100,883	109,678	63,699	107,985	124,962
R^2	0.71	0.74	0.73	0.80	0.72	0.71
	Extensive margin: LHS variable is log($P(\text{investment} > 0)$)					
	(1)	(2)	(3)	(4)	(5)	(6)
$z_{N,t}$	3.79 (1.24)	3.87 (1.21)	3.12 (2.00)	3.59 (1.14)	3.99 (1.69)	4.00 (1.13)
$CF_{it}/K_{i,t-1}$		0.029 (0.0100)				
Observations	803,659	641,173	556,011	247,648	643,913	803,659
Clusters (industries)	314	314	314	274	277	314
R^2	0.87	0.88	0.88	0.93	0.90	0.90
	Tax term: LHS variable is investment/lagged capital					
	(1)	(2)	(3)	(4)	(5)	(6)
$\frac{1-l_t z}{1-l_t}$	-1.60 (0.096)	-1.53 (0.095)	-2.00 (0.16)	-1.42 (0.13)	-2.27 (0.14)	-1.50 (0.10)
$CF_{it}/K_{i,t-1}$		0.043 (0.0023)				
Observations	637,243	633,598	426,214	211,029	510,653	631,295
Clusters (firms)	103,890	103,220	87,939	57,343	90,145	103,565
R^2	0.43	0.43	0.48	0.54	0.45	0.44
Controls	No	No	No	No	Yes	No
Industry trends	No	No	No	No	No	Yes

Notes: This table estimates regressions of the form

$$f(I_{it}, K_{i,t-1}) = \alpha_i + \beta g(z_{N,t}) + \gamma X_{it} + \delta_i + \varepsilon_{it}$$

where I_{it} is eligible investment expense and $z_{N,t}$ is the present value of a dollar of eligible investment computed at the four-digit NAICS industry level, taking into account periods of bonus depreciation. Column 2 augments the baseline specification with current period cash flow scaled by lagged capital. Column 3 focuses on the early bonus period and column 4 focuses on the later period. Column 5 controls for four-digit industry average Q for public companies and quartics in assets, sales, profit margin, and firm age. Column 6 includes quadratic time trends interacted with two-digit NAICS industry dummies. Ratios are censored at the 1 percent level. All regressions include

Figure 1 and Table 3.

5.3 Table 4 + Figure 2: robustness and depreciation-kink salience

Read:

- Table 4 shows no meaningful effect on structures investment, a placebo outcome largely ineligible for bonus.
- Net investment responds positively, suggesting the result is not merely reporting gross investment differently.
- Bonus take-up rises strongly with $z_{N,t}$, confirming that firms in high-exposure industries actually claim bonus.
- Figure 2 shows sharp bunching around Section 179 kinks, and the bunching location moves when the statutory kink moves.

Economic message: The robustness evidence supports both identification and salience: firms understand the depreciation schedule and react to its incentives.

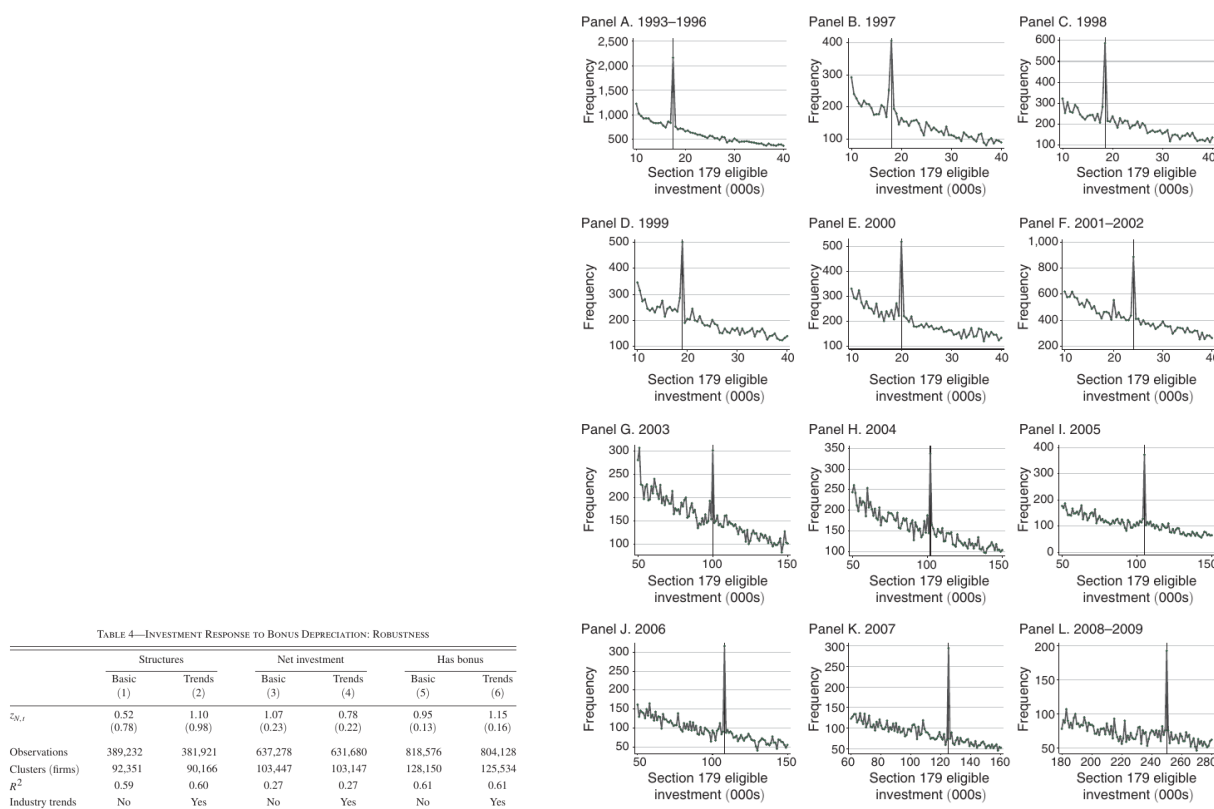


FIGURE 2. DEPRECIATION SCHEDULE SALIENCE

Notes: These figures illustrate the salience of nonlinearities in the depreciation schedule. They show sharp bunching of Section 179 eligible investment around the depreciation schedule kink from 1993 through 2009. Each plot is a histogram of eligible investment in our sample in the region of the maximum deduction for a year or group of years. Each dot represents the number of firms in a \$250 bin. The vertical lines correspond to the kink point for that year or group of years.

Source: Authors' calculations

Table 4 and Figure 2.

5.4 Table 5 + Table 6: substitution, financing, and constrained firms

Read:

- Table 5 finds positive responses in rents, payroll, and especially debt; the dividend-payer indicator falls.
- The investment coefficient in Table 5 remains large even after adding lagged policy exposure, while the

lagged coefficient is negative and insignificant.

- Table 6 shows the strongest heterogeneity: small firms have a coefficient of 6.29 versus 3.22 for big firms; low-cash firms have 7.21 versus 2.76 for high-cash firms.
- Non-dividend payers also respond more than dividend payers, 5.98 versus 3.67.

Economic message: Bonus does not merely reallocate spending across narrow categories. The response is strongest where finance or liquidity constraints are likely to bind.

TABLE 5—SUBSTITUTION MARGINS AND EXTERNAL FINANCE

	Δ Rents (1)	Δ Payroll (2)	Δ Debt (3)	Payer? (4)	Investment (5)
$z_{N,t}$	0.75 (0.26)	1.49 (0.20)	1.84 (0.21)	-0.36 (0.089)	4.22 (0.62)
$z_{N,t-2}$					-0.86 (0.69)
Observations	574,305	624,918	642,546	818,576	476,734
Clusters (firms)	98,443	102,043	103,868	128,150	84,777
R^2	0.18	0.23	0.20	0.68	0.76

Notes: This table estimates regressions of the form

$$Y_{it} = \alpha_i + \beta z_{N,t} + \gamma X_{it} + \delta_i + \varepsilon_{it}$$

where Y_{it} equals the difference in the logarithm of the dependent variable in columns 1 through 3. In column 4, the dependent variable is an indicator for positive dividend payments. $z_{N,t}$ is the present value of a dollar of eligible investment computed at the four-digit NAICS industry level, taking into account periods of bonus depreciation. Column 5 includes contemporaneous and twice lagged $z_{N,t}$. All regressions include firm and year fixed effects. Standard errors clustered at the firm level are in parentheses.

Source: Authors' calculations

TABLE 6—HETEROGENEITY BY EX ANTE CONSTRAINTS

	Sales		Div payer?		Lagged cash	
	Small	Big	No	Yes	Low	High
$z_{N,t}$	6.29 (1.21)	3.22 (0.76)	5.98 (0.88)	3.67 (0.97)	7.21 (1.38)	2.76 (0.88)
Equality test	$p = 0.030$		$p = 0.079$		$p = 0.000$	
Observations	177,620	255,266	274,809	127,523	176,893	180,933
Clusters (firms)	29,618	29,637	39,195	12,543	45,824	48,936
R^2	0.44	0.76	0.69	0.80	0.81	0.76

Notes: This table estimates regressions from the baseline intensive margin specification presented in Table 3. We split the sample based on pre-policy markers of financial constraints. For the size splits, we divide the sample into deciles based on the mean value of sales, with the mean taken over years 1998 through 2000. Small firms fall into the bottom three deciles and big firms fall into the top three deciles. For the dividend payer split, we divide the sample based on whether the firm paid a dividend in any of the three years from 1998 through 2000. The dividend split only includes C corporations. The lagged cash split is based on lagged residuals from a regression of liquid assets on a ten-piece spline in total assets and fixed effects for four-digit industry, year, and corporate form. The comparison is between the top three and bottom three deciles of these lagged residuals. All regressions include firm and year fixed effects. Standard errors clustered at the firm level are in parentheses.

Source: Authors' calculations

Table 5 and Table 6.

5.5 Figure 3 + Table 7: tax position and immediacy

Read:

- Figure 3 shows bunching at Section 179 kinks for firms with positive current taxable income, but little bunching for firms with negative current taxable income.
- Firms with larger lagged loss carryforward stocks display less excess mass at the kink.
- Table 7 translates the same idea into the bonus setting: taxable firms respond strongly to bonus, while nontaxable firms have weak or insignificant responses.
- Column 7 shows that even among taxable firms, larger loss carryforwards reduce the response.

Economic message: The same statutory deduction matters much more when it creates cash today. Delayed tax benefits are much less effective.

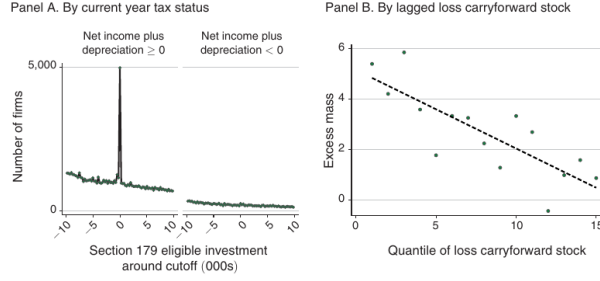


FIGURE 3. BUNCHING BEHAVIOR AND TAX INCENTIVES

Notes: These figures illustrate how bunching behavior responds to tax incentives. Firms bunch less when eligible investment provides less cash back now. Panel A splits the sample based on whether firm net income before depreciation is greater than or less than zero. Firms with net income before depreciation less than zero can carry back or forward deductions from eligible investment, but have no more current taxable income to shield. Panel B groups firms with current year taxable income based on the size of their prior loss carryforward stocks. The x-axis measures increasing loss carryforward stocks relative to current year income. The y-axis measures the excess mass at the kink point for that group. Firms with more alternative tax shields find investment a less useful tax shield and therefore bunch less. Excess mass ratios are computed using the algorithm and code in Chetty et al. (2011).

Source: Authors' calculations

TABLE 7—HETEROGENEITY BY TAX POSITION

	LHS variable is log(investment)						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Taxable $\times z_{N,t}$	3.83 (0.79)	3.08 (0.93)	1.95 (0.92)	6.43 (1.46)	4.32 (0.96)	4.15 (0.82)	
$z_{N,t}$	-0.15 (0.90)	0.60 (1.05)	0.38 (1.06)	-3.03 (1.55)	-0.69 (1.15)	0.88 (0.94)	5.68 (1.70)
Medium LCF $\times z_{N,t}$							-2.56 (1.46)
High LCF $\times z_{N,t}$							-3.70 (1.55)
CF_{it}/K_{it-1}		0.14 (0.028)					
Taxable $\times CF_{it}/K_{it-1}$		0.27 (0.035)					
Observations	735,341	580,422	514,035	221,306	585,914	722,262	119,628
Clusters (firms)	128,001	100,883	109,678	63,699	107,985	124,962	40,282
R^2	0.71	0.74	0.74	0.80	0.73	0.72	0.84
Controls	No	No	No	No	Yes	No	No
Industry trends	No	No	No	No	No	Yes	No

Notes: This table estimates regressions from each intensive margin specification, in columns 1 through 6, presented in Table 3. For each firm year, we generate an indicator based on whether a firm is in taxable position prior to depreciation expense. We fully interact this indicator with all controls and the time effects. Column 7 splits taxable firms into three groups based on the size of their lagged loss carryforward stocks relative to net income before depreciation. We interact these group indicators with $z_{N,t}$ and the time effects. Only firms with nonzero stocks of lagged loss carryforwards are included. All regressions include firm and year fixed effects. Standard errors clustered at the firm level are in parentheses.

Source: Authors' calculations

Figure 3 and Table 7.

5.6 Figure 4: comparison to past work and firm-size heterogeneity

Read:

- Panel A compares the paper's tax-term elasticity to past studies. The paper's equal-weighted estimate is larger than many prior estimates.
- Panel B shows why: smaller firms have much larger elasticities than large firms.
- The top 5% of firms account for about 62% of eligible investment, so investment-weighted elasticities are lower than equal-weighted elasticities.
- The investment-weighted elasticity is 2.89. The elasticity for the largest-firm group alone is 2.27, so including smaller firms raises the investment-weighted estimate by about 27% relative to using only the largest firms.

Economic message: Prior public-firm samples understate the policy response because they miss many smaller, more responsive firms.

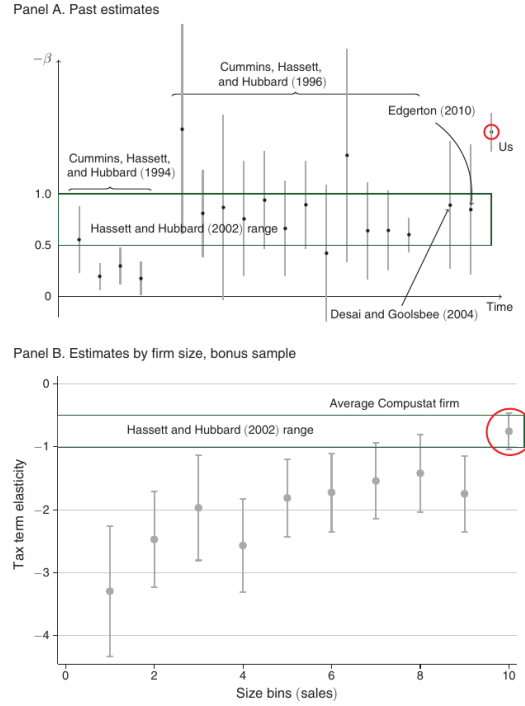


FIGURE 4. PAST ESTIMATES AND HETEROGENEOUS EFFECTS BY FIRM SIZE

Notes: These figures plot coefficients and confidence bands from tax term specifications (see the third row of Table 3) for past studies of tax reforms and our sample. Panel B splits the sample into deciles based on mean pre-policy sales. The average firm in Compustat during this time period falls in the tenth size bin (with sales equal to \$1.8 billion), which coincides with the Hassett and Hubbard (2002) survey range of user cost elasticity estimates (-0.5 to -1).

Sources: Panel A: Studies cited in online Appendix Table B8. Panel B: Authors' calculations.

6 Short summary

- Bonus depreciation accelerates deductions, lowering the effective price of eligible equipment investment.
- The empirical strategy compares industries whose normal depreciation schedules make bonus more or less valuable.
- The baseline response is large: eligible investment rises by about 10.4% in the first episode and 16.9% in the second.
- Parallel-trend graphs, structures placebos, net-investment tests, and bonus take-up tests support the design.
- Small firms, non-dividend payers, and low-cash firms respond most strongly.
- Taxable firms respond strongly; loss firms respond weakly, even though loss firms can carry deductions forward.
- Section 179 bunching provides an independent salience and immediacy test.
- The most plausible interpretation is that financial frictions, high effective discount rates, or fixed adjustment costs amplify the investment response.

7 Conclusion

7.1 Core conclusion

Temporary depreciation incentives can have large effects on equipment investment, but the effects are heterogeneous. Firms that are smaller, more financially constrained, or able to use deductions immediately respond much more strongly.

7.2 Economic intuition

Bonus depreciation looks small in a frictionless present-value calculation because it only changes the timing of deductions. But for firms that value cash today much more than cash later, the first-year tax saving is economically important. This makes immediate expensing especially powerful for constrained firms and much weaker for firms in tax-loss positions.

7.3 Takeaway

Investment stimulus works best when it targets the firms for which the tax benefit is immediate and financially valuable. A representative-firm user-cost model misses the key lesson: the distribution of firms matters for both empirical estimates and policy design.